Comparing Problem-Specific and Mean Hidden-State Interventions for Context-Reconstructed Reasoning

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LLMs can fail after a context change

Original (ORG)

A man fell from a 100-rung ladder, but was unhurt. How?



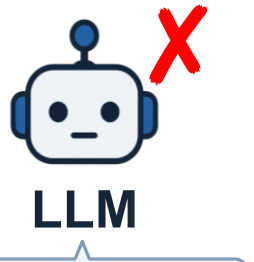


Correct answer: the lowest rung

Context-Reconstructed (CR)

A man fell from a 50-story building, but was unhurt. How?





Correct answer: the lowest floor

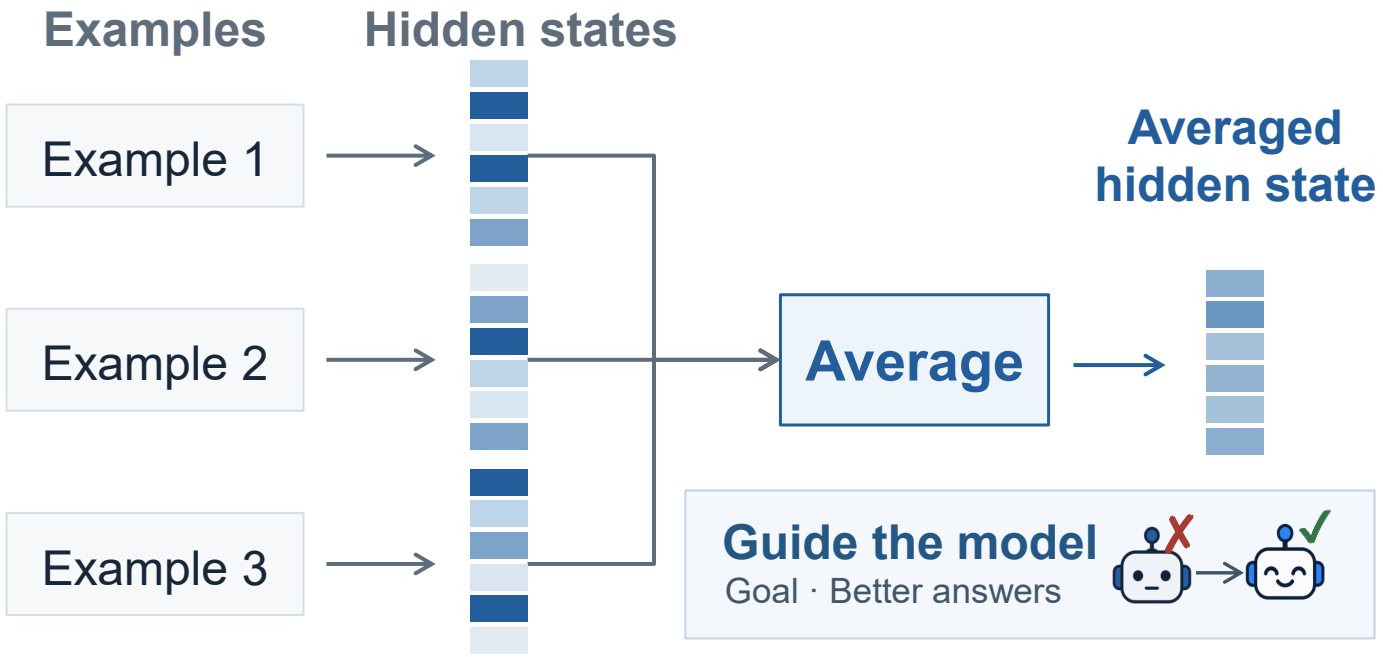
Different contexts, the same underlying solution structure.

Related Work: Interventions with averaged states

Hidden states

A list of numbers
for one token at one layer

- Changing hidden states can alter model behavior. [6–8]
- Many interventions use average directions. [6–8]



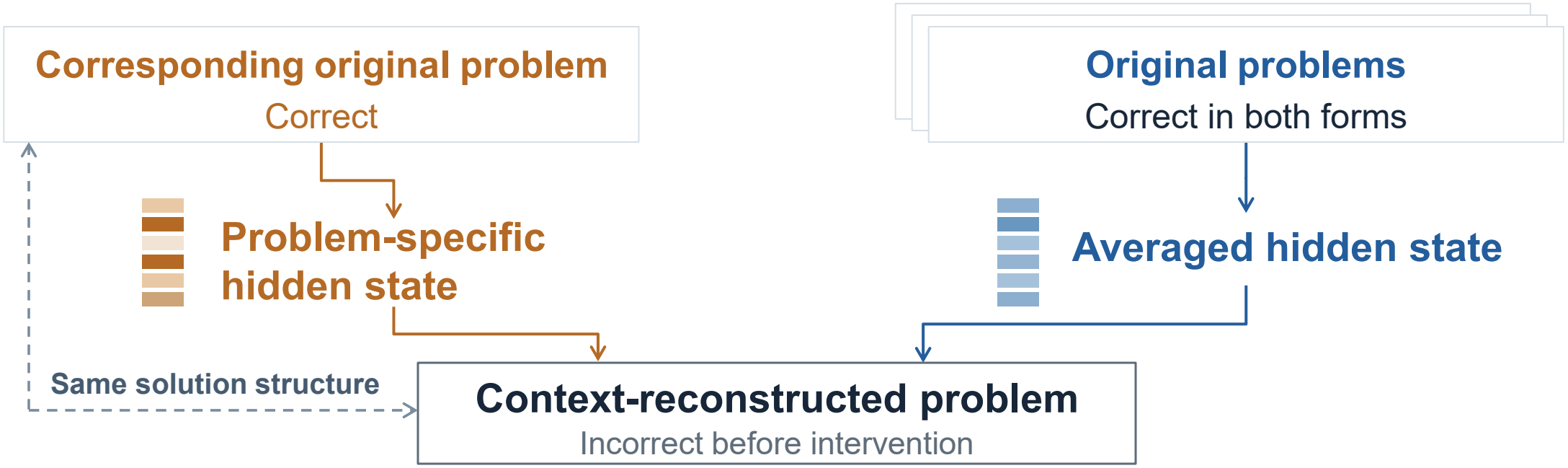
Potential limitation

Averaging may weaken problem-specific information.

Li et al. (2023), Liu et al. (2024), Todd et al. (2024)

Research Question

Does a problem-specific hidden state recover reasoning failures more effectively than an averaged hidden state?



Main finding

Problem-specific hidden states give higher peaks in all four models

Problem pairs and answer evaluation

BrainTeaser Sentence Puzzle (BTSP) — **209 problem pairs**

Four-choice lateral-thinking word puzzles

Original (ORG) → Different context, **Same solution structure** → **Context-Reconstructed (CR)**

Input

Question +
four answer options

Each model

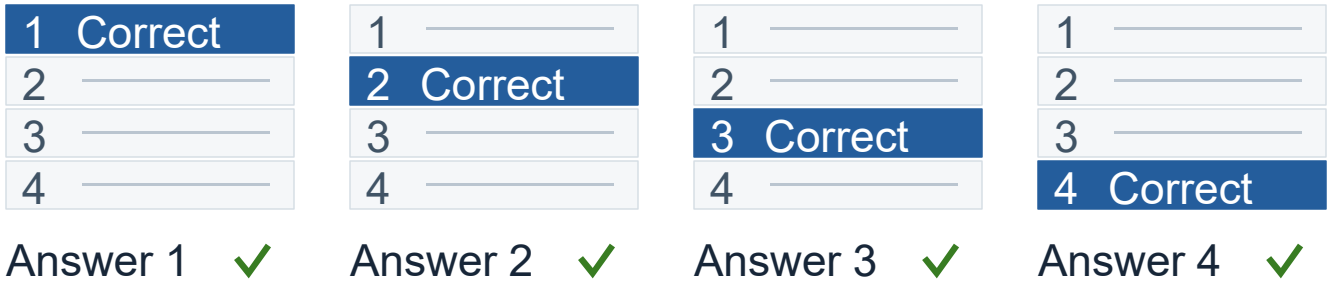
- Gemma 2 9B IT
- Gemma 3 12B IT
- Gemma 4 E2B IT
- Gemma 4 E4B IT

Output

Answer
number 1–4

Zero-shot. Select the highest candidate log-likelihood.

Same correct option, four positions (ORG / CR separately)



All 4 answers correct

Problem correct

Any error: incorrect

Failures after a context change

Letter order: ORG / CR C = Correct I = Incorrect

CI group · Targets for recovery

CI : ORG **Correct** ／ CR **Incorrect**

CC group · Sources for averaging

CC : ORG **Correct** ／ CR **Correct**

Problem pairs counted separately for each model

Model	**CI group** **CR targets**	**CC group** **ORG sources**
Gemma 2 9B IT	**40**	113
Gemma 3 12B IT	**38**	94
Gemma 4 E2B IT	**35**	11
Gemma 4 E4B IT	**32**	99

Choosing hidden states

Problem-specific (Pair-CI)

Hidden state from the corresponding ORG problem

Averaged (Mean-CC)

Average of ORG hidden states from the CC group

Additional comparisons	ORG source	Comparison purpose
Mean-CI	Averaged ORG hidden state from other CI problems	Source-group membership
Surface-similar-CI / Surface-similar-CC	ORG with highest TF-IDF similarity to CR	Lexical similarity
Random-CI / Random-CC	One randomly selected ORG	Arbitrary source choice

CI-source controls use Gemma 4 models and exclude the corresponding ORG problem.

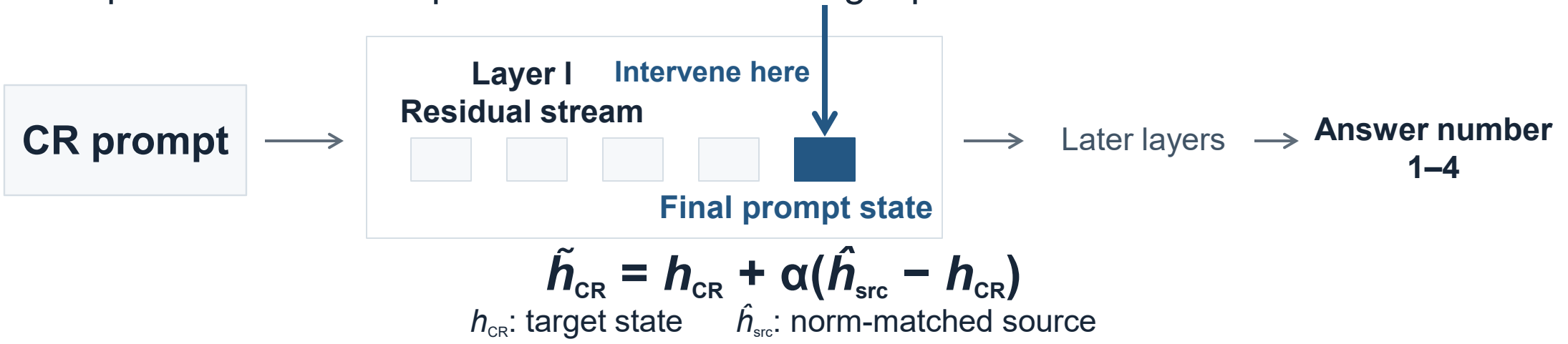
Hidden-state intervention

Source hidden state

Average hidden states over four option-slot variants per ORG

Norm-matched source

Match the norm of the target problem's hidden state



$$\tilde{h}_{CR} = h_{CR} + \alpha(\hat{h}_{src} - h_{CR})$$

h_{CR}: target state $\hat{h}_{src}$: norm-matched source

Main comparison

Full replacement (α = 1.0) for every source

Test every layer, one at a time

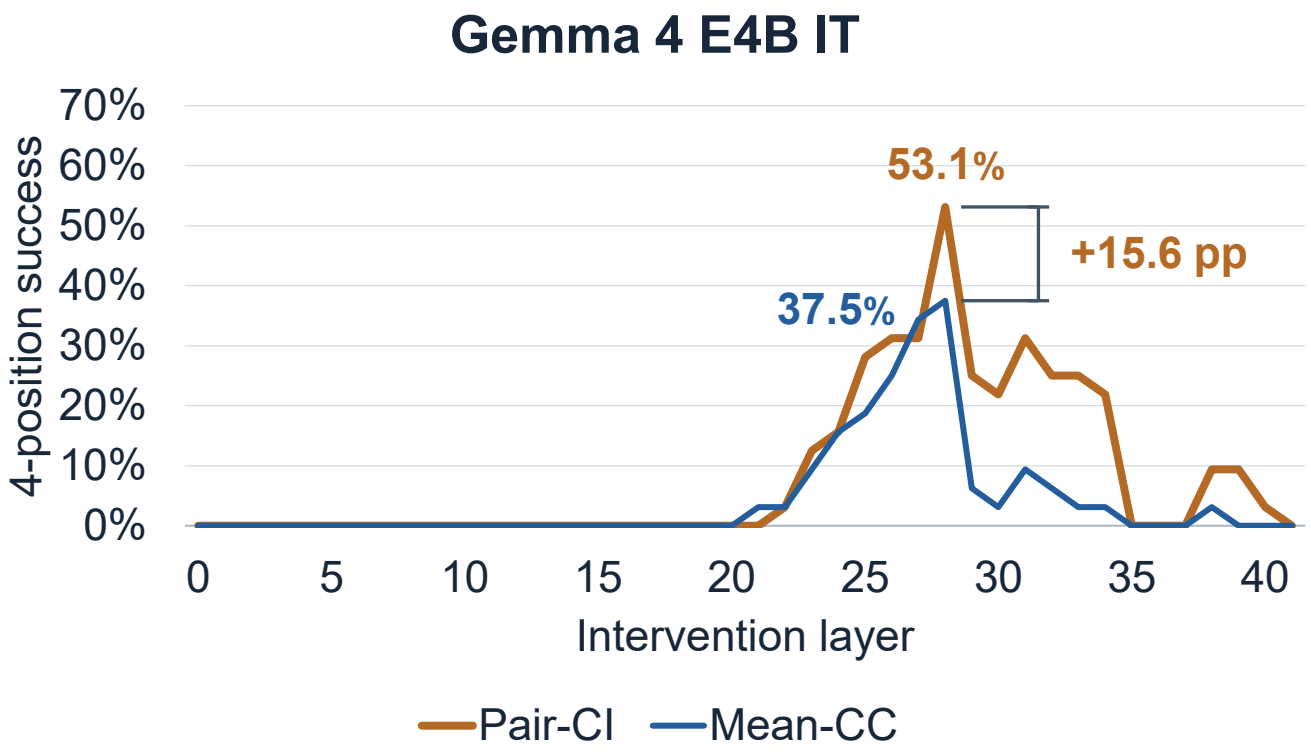
Evaluation

Can the model now solve CR problems it previously failed?

4-position success rate = % of CI group CR targets correct in all four positions

Higher peak recovery in all four models

α = 1.0 Gaps in percentage points (pp)



Peak recovery

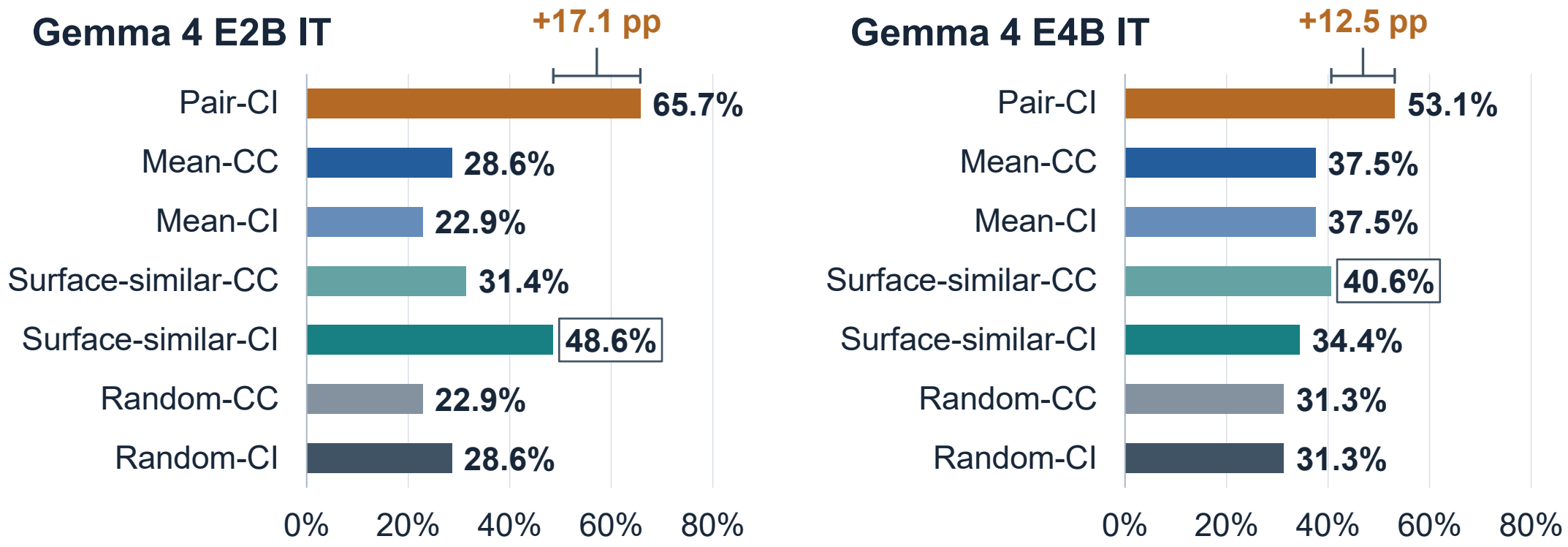
Model	Pair-CI	Mean-CC
Gemma 2 9B IT	**22.5%**	15.0%
Gemma 3 12B IT	**15.8%**	7.9%
Gemma 4 E2B IT	**65.7%**	28.6%
Gemma 4 E4B IT	**53.1%**	37.5%

Peak over layers

Averaging across problems may weaken useful information.

Other sources give lower peaks

Gaps: Pair-CI vs best alternative; peaks over layers at α = 1.0.

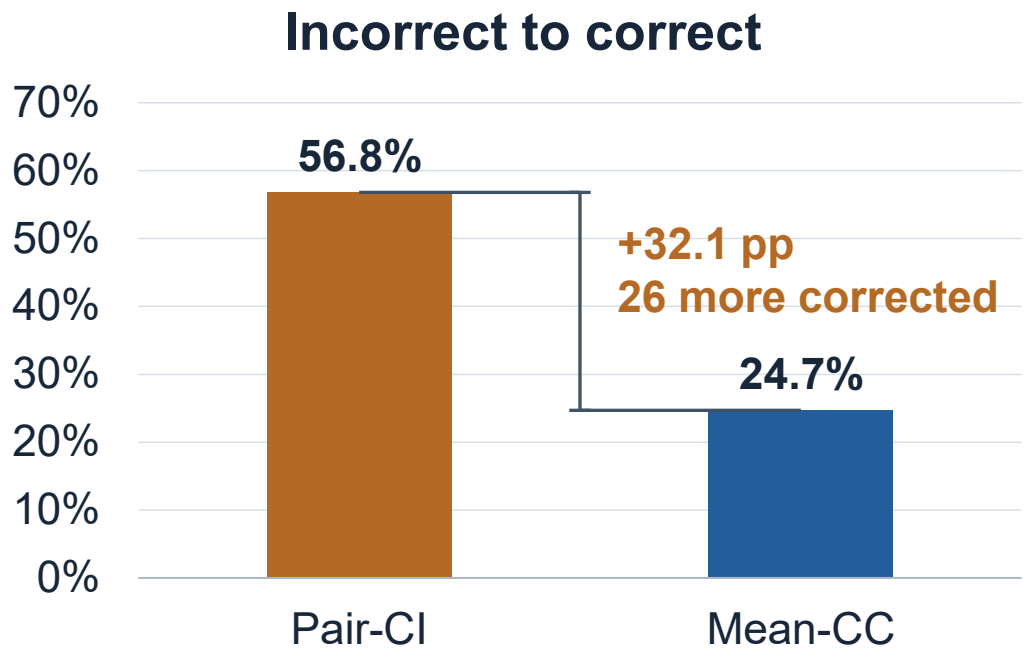


The tested group and lexical controls do not reproduce the Pair-CI peak.

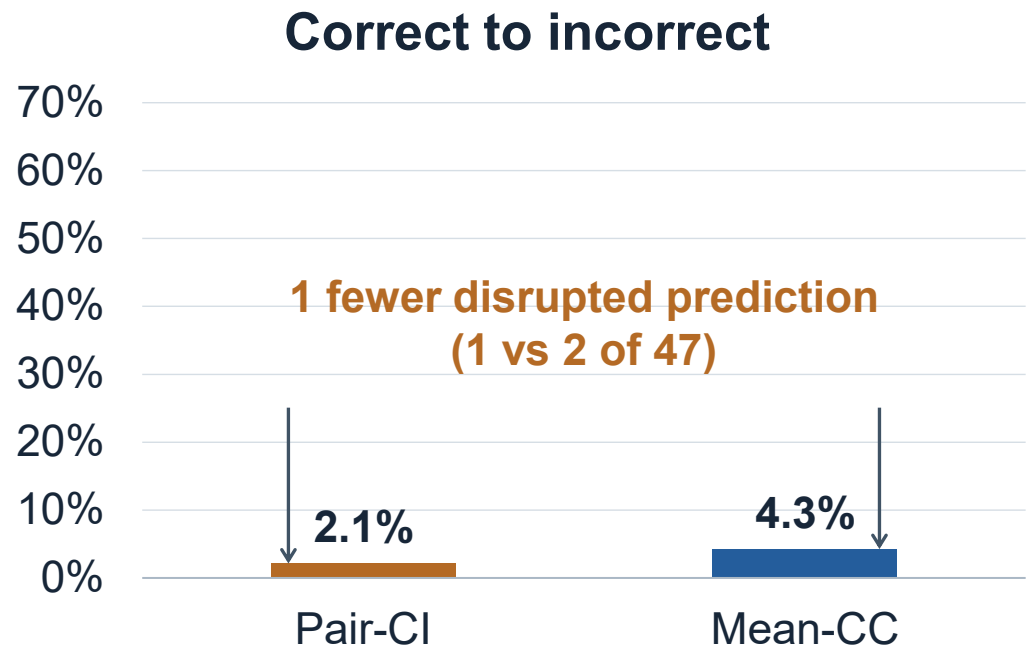
Correcting errors with little disruption

Gemma 4 E4B IT, layer 28, α = 1.0

81 initially incorrect predictions



47 initially correct predictions



At this layer, Pair-CI corrects more errors with little disruption.

Limitations

Meaning	Hidden states may contain clues to the solution. Their meaning and the full solution structure are unclear.
Scope	**Only** one dataset and one failure group (CI). Four models, one prompt position, one layer at a time.
Peak estimates	Descriptive maxima across the tested layers. No held-out selection or uncertainty estimates.

Answer to the research question

Does a **problem-specific hidden state** recover reasoning failures more effectively than an **averaged hidden state**?

Problem-specific states give higher peak recovery.

All four tested models, α = 1.0 (Pair-CI vs Mean-CC)

Pair-CI on E4B, layer 28: more error correction
with little disruption of correct predictions.

Next Identify the useful information and test broader reasoning tasks.